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WHY SHOULD WE USE AUTO COMMIT = FALSE?

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By default, JDBC uses:

Auto Commit = true

This means every SQL statement is saved immediately after it executes successfully.

For a single query, this is usually fine.

Example:

```
INSERT INTO students VALUES(...);
```

The record is inserted and saved instantly.

Why is this a problem?

Many real-world tasks require multiple SQL queries to complete one operation.

Example:

A bank money transfer.

Step 1:

Deduct ₹1000 from Account A

Step 2:

Add ₹1000 to Account B

These two queries belong to ONE transaction.

If Auto Commit is ON:

Query 1 → Saved ✓

Query 2 → Error ✗

Result:

Account A loses ₹1000.
Account B receives nothing.

The database now contains incorrect and inconsistent data.

Solution

Disable Auto Commit.

```
con.setAutoCommit(false);
```

Now JDBC waits instead of saving each query immediately.

If every query succeeds:

```
con.commit();
```

All changes are saved together.

If any query fails:

```
con.rollback();
```

All changes are cancelled.

This keeps the database correct and consistent.

WHAT HAPPENS IF WE DON'T USE IT?

If we leave Auto Commit ON:

- ✓ Every query is saved immediately.
- ✓ We cannot group multiple queries into one transaction.
- ✓ If one query fails, previous successful queries remain saved.
- ✓ This may leave incomplete or incorrect data.
- ✓ Data consistency can be lost.

Example:

UPDATE Account A ✓ Saved

UPDATE Account B ✗ Failed

Final Result:

Account A is updated.

Account B is not updated.

This creates an inconsistent database.

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BENEFITS OF AUTO COMMIT = FALSE

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- ✓ Groups multiple SQL statements into one transaction.
- ✓ Prevents partial updates.
- ✓ Keeps data accurate and consistent.
- ✓ Gives complete control over when data is saved.
- ✓ Allows rollback() if an error occurs.
- ✓ Improves performance during batch processing because the database commits once instead of after every query.

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WHEN SHOULD WE USE IT?

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Use Auto Commit = false when:

- Money transfers
 - Batch inserts
 - Online shopping orders
 - Employee salary processing
 - Booking systems
 - Inventory management
 - Any operation involving multiple related SQL queries
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SIMPLE RULE TO REMEMBER

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Auto Commit = true

Every query is saved immediately.

Auto Commit = false

Wait...

Execute all queries...

If all succeed:

commit();

If any fail:

rollback();

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INTERVIEW ANSWER

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Q: Why do we use Auto Commit = false?

Answer:

We use Auto Commit = false to group multiple SQL statements into a single transaction. This ensures that either all queries are saved together using commit(), or all changes are cancelled using rollback() if an error occurs. It prevents partial updates and keeps the database consistent.